STAT238 Lecture 1 Introduction to Bayesian Statistics

1 Introduction to Bayesian Statistics

1.1 Bayesian Statistics 是什么

根据 "Probability Theory – the Logic of Science" 书中的表述,

$$\text{Bayesian Statistics} = \text{Probability Theory}$$

1.2 Bayesian Statistics 如何运作

核心直觉: Probability as Belief

- 当我们无法完全确定一个命题是否为真时, 可以用一个 0 到 1 之间的概率来表示我们对其真实性的信念强度
- 概率的作用: 在不完全信息下, 对 "命题为真" 的可信程度进行定量表达

概率推断的两个基本步骤:

Step 1: Modeling Step

将已有/假设信息信息转化为 numerical assignments for the probabilities

Step 2: Inference Step

给定模型后, 使用 rule of probability 计算我们关心的事件的概率

Bayesian 语境中的表述:

- 未知真值通常表示为变量: $\Theta \in \mathcal{A}$ for some set $\mathcal{A}$
- Inference step 涉及到计算 posterior probability:

$$\mathbb{P}\{\Theta \in \mathcal{A} \mid \text{observed data and other background information}\}$$

- Modeling step 涉及到明确
  - likelihood:

$$\mathbb{P}\{\text{observed data} \mid \Theta = \theta \text{ and background information}\}$$

  - prior model / prior distribution:

$$\mathbb{P}\{\Theta = \theta \mid \text{background information}\} \quad \text{or} \quad f_{\Theta \mid \text{backgournd information}}(\theta)$$

2 Rules of Probability

Logic

Probability 是分配给 propositions / events 的, 通常 probability 会 conditional on some information (包括已知的信息和假设的信息)

2.1 Conditional Probability

若 proposition A 的概率是基于某些 information I, 则记该条件概率为

$$\mathbb{P}(A \mid I)$$

Remark

若 information I is clear from context, 则可以将条件概率 $\mathbb{P}(A \mid I)$ 简写为 $\mathbb{P}(A)$, 但我们仍需明确记住这些概率是基于某些信息的

2.2 Rule of Probability

1. **Basic Bounds:** $0 \leq \mathbb{P}(A \mid I) \leq 1$
2. **Product Rule:** $\mathbb{P}(A \cap B \mid I) = \mathbb{P}(A \mid I)\mathbb{P}(B \mid A, I) = \mathbb{P}(B \mid I)\mathbb{P}(A \mid B, I)$
3. **Sum Rule:** 若 proposition A 可以被划分为 disjoint propositions $A_1, A_2, \ldots$, 则 $\mathbb{P}(A \mid I) = \sum_i \mathbb{P}(A_i \mid I)$

基于以上 rules, 可以推出 Bayes rule:

$$\begin{aligned}
\mathbb{P}(B \mid A, I) &= \frac{\mathbb{P}(A \mid B, I)\mathbb{P}(B \mid I)}{\mathbb{P}(A \mid I)} \\
&= \frac{\mathbb{P}(A \mid B, I)\mathbb{P}(B \mid I)}{\mathbb{P}(A \mid B, I)\mathbb{P}(B \mid I) + \mathbb{P}(A \mid B^c, I)\mathbb{P}(B^c \mid I)}
\end{aligned}$$